
When Shift Happens - Confounding Is to Blame

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Abstract

Distribution shifts create measurable gaps between in-distribution and out-of-distribution performance, and these gaps remain a central obstacle to robust machine learning. Although causal-invariant representations are often treated as the main route to robustness, recent empirical comparisons show a sharper quantitative pattern: (i) empirical risk minimization (ERM) can match or exceed state-of-the-art out-of-distribution (OOD) generalization methods, and (ii) OOD accuracy often improves when models use all available covariates rather than only the causal subset. We use theoretical analysis and empirical evidence to explain these margins through hidden confounding. Shifts in hidden confounding change observed data distributions in ways that violate assumptions used by many OOD methods. In this regime, we prove that maximizing predictive information requires environment-specific relationships in addition to invariant ones. We further show that adding proxy variables for hidden confounders improves the relevant information terms and mitigates hidden-confounding shift. Together, the results clarify why all-variable and ERM-style models can perform so strongly in practice and provide quantitatively grounded guidance for OOD algorithms and covariate selection.

1 Introduction

Generalization measures whether conclusions learned from observed data remain reliable on unseen data, and the same quantitative concern appears across scientific domains. In medicine and the social sciences, it is measured through *external validity*, where estimates from one population must transfer to another [13]; in ecology, *space-for-time substitutions* use spatial variation as empirical evidence for temporal change [60]; and in engineering, *robust control* evaluates whether performance remains stable under unmodeled disturbances [35]. Machine learning has increasingly become the modeling framework used to pursue these generalization goals at scale [100, 88, 49].

For machine learning, the operational test of generalization is performance under distribution shift between training and deployment. A model trained to predict *food stamp reciprocity* from *household attributes* in one state should retain comparable accuracy when evaluated in another state. We refer to this requirement as *out-of-distribution (OOD) generalization*: the predictor must perform well on both in-distribution (ID) data, from which training samples are drawn, and OOD test data. Over the last decade, domain generalization [56, 31, 53, 99, 6, 38, 74], domain adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94] have each proposed objectives with provable OOD guarantees under specific assumptions. However, the empirical comparison has become more nuanced: with careful model selection [30, 38, 87], standard *empirical risk minimization (ERM)* [86] is often competitive with, and sometimes stronger than, specialized OOD methods on real-world benchmarks [30, 24, 50, 57, 67, 87]. A recent benchmark across 16 real-world tabular datasets further found that using all available covariates, rather than restricting to direct causal covariates, can improve OOD performance [57]. These repeated empirical margins motivate a mechanism-level explanation of when and why broader covariate sets and ERM-style objectives win.

Distribution shifts arise naturally when samples come from multiple environments [71]. In the food-stamp example, state-specific eligibility rules induce different environment-level distributions of both covariates and outcomes. Some covariate–outcome relationships may nevertheless remain stable across environments; these are the *invariant relationships* targeted by many OOD methods [6].

Learning such invariant relationships has produced a large body of generalization guarantees [6, 59, 31, 66, 56, 98, 63]. A particularly important case is *causal invariance*, where the causal relationships in the data remain stable across environments [71, 59, 6]. The empirical difficulty is that real datasets rarely contain every relevant predictor: data collection limits, privacy constraints, and measurement error routinely leave important variables unobserved [14, 52, 20, 48]. When these missing variables confound both observed covariates and the outcome, their distributional changes create environment shifts that alter several observed quantities at once. Ignoring these hidden-confounding shifts can reduce OOD accuracy [40, 4, 85, 62] and can also make the learned covariate–outcome relationship quantitatively misleading.

Despite this practical importance, the quantitative effect of hidden confounding shifts on OOD generalization remains underexplored [4, 40, 85, 62]. This work connects theoretical information decompositions with the empirical comparisons reported in recent benchmarks [30, 24, 50, 57, 67, 87], using those comparisons to explain why certain methods and covariate sets achieve larger ID and OOD gains. Our contributions are as follows:

- We motivate confounding shift in OOD generalization from a causal perspective (§ 3) and identify the comparison pattern under which informative, non-causal covariates improve performance.
- We show that maximizing predictive information between predictions and outcomes requires explicitly learning environment-specific relationships; invariant representations remain useful, but the decomposition shows they are not sufficient by themselves (§ 4.1, § 4.2).
- We prove that variables informative about the outcome or hidden confounders improve the favorable terms in predictive information, clarifying why covariate selection changes observed accuracy margins under hidden-confounding shift (§ 4.3).
- Across real-world and synthetic experiments, we report consistent evidence that (i) hidden confounding is prevalent in tabular benchmarks, (ii) environment-specific relationships are positively aligned with ID and OOD test accuracy, and (iii) informative, non-causal covariates improve generalization (§ 5).

Overall, the analysis reframes the strong empirical performance of ERM and all-variable models as a predictable outcome of hidden-confounding shift, and it highlights the need for methods whose objectives track the information terms that actually move ID and OOD accuracy.

2 Related work

Out-of-distribution (OOD) generalization covers several evaluation settings, including domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. These settings differ in what evidence is available at training or evaluation time: domain adaptation uses unlabeled test-domain data, domain generalization uses training environment labels, federated learning aggregates data across environments subject to communication and privacy constraints, and OOD detection scores whether a sample departs from the training distribution. A common design principle is to learn invariant relationships, but empirical and theoretical comparisons increasingly show that invariance alone does not account for observed generalization margins [46, 72, 93]. In particular, Ye et al. [93] argue that feature invariance is necessary but not sufficient and that *informativeness* must also be measured. Our analysis keeps both quantities explicit and tracks how each contributes to OOD performance.

Proxy variable adjustment: When confounding variables are available during training but absent at test time, Landeiro and Culotta [40] adjust for confounding shifts and report improved classifier performance. Proxy-based causal-effect adjustment [55, 39] has also motivated domain-adaptation methods that recast unknown distribution shift as causal effect identification [4, 85]. More recent OOD work studies hidden confounding shift under overlapping confounder support [62]. Our focus is complementary: rather than only proposing an adjustment, we quantify which predictive-information terms improve when proxy variables are added and how those improvements align with OOD accuracy.

All variable models vs causal models. Recently, Nastl and Hardt [57] benchmarked covariate subsets grouped as causal, arguably causal, anti-causal, and other spurious covariates. Across 16 datasets, the all-covariate models Pareto-dominate the causal-only or arguably-causal subsets on both ID and OOD data, creating a clear empirical comparison in favor of broader covariate sets. Theoretical explanations

for this pattern remain limited, and our results provide one under hidden-confounding shift. For linear causal models, anchor regression [68] balances models that use all observed covariates against models restricted to causal covariates. We instead ask how adding non-causal but informative covariates changes the information terms that determine predictive performance. Relatedly, Eastwood et al. [21] show that unstable covariates can improve performance when they contain label information and are conditionally independent of stable covariates given the label; their method adjusts shift using unlabeled test-domain data. Yet on a medical real-world dataset not constructed for that setting [8], ERM remains competitive, matching the broader evidence from Nastl and Hardt [57]. This is also consistent with the result that maximum likelihood estimation (MLE) is minimax optimal under well-specified covariate shifts [26]. Real-world tabular benchmarks, however, are often not well-specified because hidden confounding shifts change multiple observed distributions simultaneously, which is the setting we analyze.

3 Manifestations of hidden confounding shift

This section makes the hidden-confounding comparison explicit: we enumerate the observed shifts that can arise from the same unobserved source and use causal-effect identification to show why the induced quantitative ambiguity matters for OOD prediction.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , Table 1 separates the observable comparisons between environments e and e' into label, covariate, conditional covariate, concept, and joint dataset shifts. A single shift in the unobserved distribution $\mathbb{P}(U)$ can generate any of these measured discrepancies when U causes $\mathbf{X}$, Y , or both (Figure 1). For example, $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$, so $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ together with $U \rightarrow \mathbf{X}$ implies the observed margin $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. The same accounting applies to $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$, which is why hidden confounding can appear simultaneously in multiple shift metrics.

Table 1: Observed distribution-shift metrics compared across environments e and e' .

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

Several existing OOD methods rely on causal structures among $U, \mathbf{X}, Y$ that make one of these shift metrics vanish or remain invariant. For instance, when $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), since $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$, a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to label shift, i.e., $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$, but conditional covariate distribution stays invariant i.e., $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$ because $\mathbb{P}(\mathbf{X} | Y, U) = \mathbb{P}(\mathbf{X} | Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. Similarly, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1 (b)), a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is observed due to covariate shift, i.e., $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, but the conditional distribution $\mathbb{P}(Y | \mathbf{X})$ stays invariant i.e., $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. These invariances may not hold in many scenarios because U usually causes both $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In this case, when $\mathbb{P}(U)$ shifts between environments, label shift, covariate shift, conditional covariate shift, and concept shift can all occur simultaneously. Nevertheless, confounding shift induces an invariance: $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$ referred to as *stable confounding shift* [85, 4]. Since U is unobserved, this invariance is absent from the observed data.

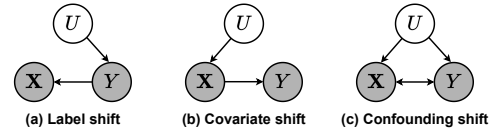


Figure 1: Causal graphs showing which hidden-confounding structures generate each observed shift pattern. (a) Label shift: $U \rightarrow Y \rightarrow \mathbf{X}$. (b) Covariate shift: $U \rightarrow \mathbf{X} \rightarrow Y$. (c) Confounding shift: $U \rightarrow \mathbf{X}$ and $U \rightarrow Y$, and $\mathbf{X} \rightarrow Y$.

Information-theoretic measures: Following Federici et al. [22], we report mutual-information analogues of the distribution shifts listed above. The quantities $I(\mathbf{X}; E)$, $I(Y; E)$, $I(\mathbf{X}; E | Y)$, $I(Y; E | \mathbf{X})$, $I(\mathbf{X}, Y; E)$, with E denoting the environment, measure shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$ respectively. Label-shift methods effectively assume $I(\mathbf{X}; E | Y) = 0$, whereas covariate-shift methods assume $I(Y; E | \mathbf{X}) = 0$. Hidden confounding is quantitatively harder because both conditional quantities can be nonzero in the same dataset.

Challenges with hidden confounding shift: Figure 2 gives a controlled quantitative example of how hidden confounding changes the learned slope and OOD error. The outcome Y depends on X and a hidden confounder U , and environment-specific shifts in the mean of U , denoted by μ_e , define the

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

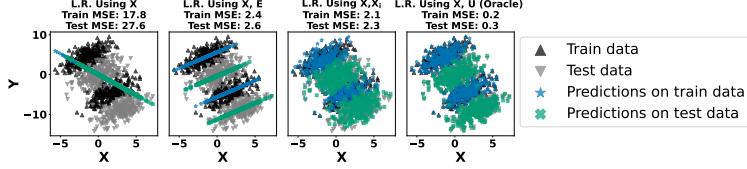


Figure 2: Four linear regression (L.R.) comparisons under hidden-confounding shift with minimal environment overlap (distant μ_e). (i) The X -only model learns the wrong relationship with Y and exhibits Simpson’s paradox. (ii) Adding environment-specific summaries of X , denoted by E , recovers the correct direction but remains representation-limited. (iii) Adding an informative proxy X_i for U improves OOD generalization. (iv) The oracle model uses both X and U and gives the best-possible MSE reference.

environments. A linear regression model trained only on observed covariate X learns the wrong X – Y relationship, as shown in the first subplot. This is Simpson’s paradox [73]: within each environment Y increases with X , yet the pooled model estimates the opposite trend. We observe the same qualitative failure for OOD methods such as IRM [6], VREX [38], and Group DRO [69]. Section 4 shows theoretically that the performance-improving strategy is to learn environment-specific relationships, which recovers the correct X – Y direction in the second subplot. Environment-specific summary statistics of X —mean, standard deviation, and quantiles—act as proxies for confounding adjustment, paralleling the *backdoor adjustment* criterion in causal effect estimation [58]. The second subplot also shows that summaries from observed covariates alone can be representationally limited; as quantified in § 4.3, additional informative covariates that proxy for U further improve generalization, as in the third subplot. The oracle model trained on X and U provides the best-possible mean squared error for comparison.

4 Impact of hidden confounding shifts on OOD generalization

This section quantifies how hidden-confounding shifts affect the objectives and metrics used in OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ be the predicted label for an input $\mathbf{X}$, where f is a classifier and ϕ is a feature extractor or transformation function. $H(X) = -\mathbb{E}_X[\log(\mathbb{P}(X))]$ denotes entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ denotes mutual information. The risk of the predictor $(f \circ \phi)$ in an environment e is defined as: $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$ where $\ell(\cdot, \cdot)$ is a loss function. The OOD target is a predictor $(f \circ \phi)$ with strong ID and OOD performance. Robust optimization methods [7, 9, 19, 75, 69] minimize the worst training-environment risk $\mathcal{R}^{\text{Rob}}(f \circ \phi) = \max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$, where $\mathcal{E}_{tr}$ is the set of training environments. Variance risk extrapolation (VREX) [38] minimizes $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$, with hyperparameter β . Invariant risk minimization (IRM) [6] minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to f being simultaneously optimal across environments. ERM-based methods pool all training environments and minimize empirical risk on the pooled data [6, 38]. That is, ERM minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$, where $\mathcal{D}^e$ is the observed data in environment e .

To compare these objectives on a shared scale, we analyze predictive information $I(Y; \hat{Y})$ with $\hat{Y} = (f \circ \phi)(\mathbf{X})$. This quantity measures how much the prediction reduces uncertainty about the true label, so larger values correspond to a more informative predictor. We decompose $I(Y; \hat{Y})$ into terms for *informativeness*, *invariance*, shift, and residual error, making clear which terms should increase or decrease for better OOD accuracy [93].

Definition 4.1 (Informativeness and Conditional Informativeness). *The informativeness of features $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informativeness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as $\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

Conditional informativeness is the within-environment evidence that $\phi(\mathbf{X})$ carries about Y . Reducing CINF removes predictive signal, whereas increasing it can improve accuracy when E is not encoding sensitive information. In fairness-sensitive settings, the same increase may be undesirable because $\phi(\mathbf{X})$ can exploit biased within-environment associations; one can reduce such dependence by minimizing $I(\phi(\mathbf{X}); E)$, but the decomposition below shows the predictive-performance cost of doing so. In the non-sensitive setting studied here, we therefore treat higher CINF as a favorable quantity and test this framing empirically in § 4.1.

Definition 4.2 (Variation and Invariance). For a given label Y , the variation in the features $\phi(\mathbf{X})$ across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e., $\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$

Invariance quantifies label-conditional consistency of $\phi(\mathbf{X})$ across environments. Lower invariance preserves more environment-specific signal and can overfit to training domains, while higher invariance removes dependencies that may not transfer. Extending the information-theoretic measures from § 3, we use $I(\phi(\mathbf{X}); E)$ and $I(Y; E | \phi(\mathbf{X}))$ to quantify *feature shift* and *concept shift* respectively. Depending on the context, we use the term *concept shift* to denote either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$. The key quantitative point of § 4.2 is that invariance is not the whole objective under hidden confounding: maximizing it alone can leave predictive information on the table.

4.1 A general decomposition of predictive information

We begin with a few causal-graph preliminaries. A causal graph $\mathcal{G}$ consists of nodes representing random variables, and directed edges indicating direct causal influences between nodes. A *path* between two nodes X_i and X_j is a sequence of unique nodes connected by edges. A *directed path* from X_i to X_j with $i < j$ is one where all arrows point toward X_j , i.e., $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. In such a directed path, X_i is called the *parent* of X_{i+1} , X_{i+1} is the *child* of X_i , X_i an *ancestor* of X_j , and X_j is a *descendant* of X_i . Paths decompose into three fundamental structures: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$, and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In both chains and forks, X_i and X_k are marginally dependent yet become conditionally independent upon conditioning on the intermediate node X_j , i.e. $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In a collider, X_i and X_k are marginally independent but become conditionally dependent when conditioning on X_j or any of its descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k | X_j$. A path between X_i and X_k is said to be *blocked* by a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$. If all paths from X_i to X_k are blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k | \mathcal{S}$. A path is open if it is not blocked.

Now consider the predictive information $I(Y; \hat{Y})$, where the predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ are based on a learned representations $\phi(\mathbf{X})$. We model the underlying causal relationships among $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$ and E as shown in Figure 3. Here, U is a hidden confounding variable. Given either $\mathbf{X}$ or $\phi(\mathbf{X})$, $\hat{Y}$ is redundant for reasoning about Y . That is, $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})$. E denotes an environment variable that captures shifts in $\mathbb{P}(U)$. That is, between any two environments, there is a shift in $\mathbb{P}(U)$. We first present a more general decomposition of predictive information without explicit consideration of how U influences $\mathbf{X}, Y$. A formal treatment of U 's influence on $\mathbf{X}$ and Y is presented in § 4.2.

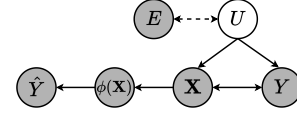


Figure 3: Bi-directed arrow between $\mathbf{X}$ and Y indicate that some covariates of $\mathbf{X}$ can cause Y , and some may be caused by Y .

Proposition 4.1. For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

Proofs are presented in Appendix § A. Existing methods can be viewed as methods that explicitly minimize or maximize certain terms in Equation (1). For instance, IRM [6] enforces that a fixed classifier f remains the same across different environments. This is directly related to maximizing $-I(\phi(\mathbf{X}); E|Y)$ because if $\phi(\mathbf{X})$ were to carry extra environment-specific information conditional on Y , the fixed classifier would no longer be optimal in all environments. DANN [23] and the independence criterion for fair classification [22] aims to minimize environment-specific information in $\phi(\mathbf{X})$ by minimizing $I(\phi(\mathbf{X}); E)$. In the context of domain adaptation, certain properties of

features are learned invariant to domains [82, 90]. While minimizing $I(\phi(\mathbf{X}); E)$ may be good for fair classification, it may degrade predictive performance [98, 33, 22]. This can be seen from Equation (1), where minimizing $I(\phi(\mathbf{X}); E)$ can reduce $I(Y; \hat{Y})$. In CDAN [51], the objective is to obtain $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$. By enforcing this equality, E becomes independent of the pair $(\phi(\mathbf{X}), Y)$. Consequently, $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$. Thus, although CDAN does not explicitly include $I(Y; E | \phi(\mathbf{X}))$ in its loss, its joint-distribution alignment objective effectively drives this conditional mutual information to zero. While $I(Y; E)$ is a constant, there exist methods that are robust to shifts in the distribution of labels [69, 95].

Equation (1) deliberately leaves unspecified how U influences $\mathbf{X}$ and Y . Once that causal role is specified, the relative importance of the terms changes and the preferred objective can be stated more sharply. That is, if $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, suggesting that minimizing $I(\phi(\mathbf{X}); E | Y)$ is a principled objective. Similarly, if $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, thereby motivating the minimization of $I(Y; E | \phi(\mathbf{X}))$. However, if $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, the interaction between the terms in Equation (1) becomes non-trivial, and it remains unclear what constitutes an ideal strategy for addressing them. We answer this question in the next section.

4.2 Predictive information under hidden confounding shift

To quantify how a hidden confounder U that causes both $\mathbf{X}$ and Y changes predictive information, we compare two common causal directions between observed covariates and labels: (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. When $\mathbf{X} \rightarrow Y$, the inequalities in (2) follow because conditioning on Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider Y , adding information flow through Y . In contrast, conditioning on $\phi(\mathbf{X})$ partially blocks information flow from Y to E at $\mathbf{X}$ when $\phi(\mathbf{X})$ retains information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

Similarly, when $Y \rightarrow \mathbf{X}$, we obtain the inequalities in (3) because conditioning on Y blocks the information flow from $\phi(\mathbf{X})$ to E at the node Y and conditioning on $\phi(\mathbf{X})$ opens the path $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at the node $\mathbf{X}$ because $\phi(\mathbf{X})$ is the child of $\mathbf{X}$ and conditioning on the child of a collider opens a path via that collider [58] (recall the preliminaries in § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

Compared to the general decomposition in (1), the inequalities in (2), (3) show that the shift terms cancel in these hidden-confounding cases, yielding the sharper decomposition below.

Proposition 4.2. *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

In Proposition 4.2, we tackle the two cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ separately. However, the causal relationship $\mathbf{X} \rightarrow Y$ is more prevalent in real-world tabular prediction tasks. For instance, 11 out of 16 datasets considered in recent benchmark studies [57, 50, 24] follow the causal structure $\mathbf{X} \rightarrow Y$. In the remaining 5 out of 16 datasets, the number of covariates that are caused by Y is much smaller than the number of covariates that cause Y . We present the key takeaway from Proposition 4.2 below.

Compared to the decomposition in (1), the decomposition in (4) (which is obtained for the special cases: (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$, (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$) is free from the terms: *variation, label shift, feature shift, and concept shift*. That is, under hidden confounding shift, when either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, the predictive information can be maximized by maximizing the difference: *conditional informativeness* – *residual*.

Maximizing conditional informativeness: In a very recent work, Prashant et al. [62] proposed a mixture of experts (MoE) model for OOD generalization under hidden confounding shift, where

each expert corresponds to a specific hidden confounder assignment. That is, each expert focuses on maximizing the performance within the environment corresponding to a particular value of the hidden confounder. Thus, learning MoE models aligns with the goal of maximizing the conditional informativeness $I(\phi(\mathbf{X}); Y|E)$. Equation (4) provides theoretical justification for the use of MoE-type models. Beyond supporting such models, the Equation (4) also highlights the need for methods that operate under more general confounding shift settings. For instance, the method proposed by Prashant et al. [62] assumes overlapping confounding support i.e., $\text{supp}(\mathbb{P}(U|\mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U|\mathcal{E}_{tr}))$, and the proxy variables to be discrete-valued.

The results in Figure 2 illustrate that, in a simple setting without confounding overlap and with a continuous-valued proxy, a linear regression model can successfully recover the correct causal relationship between observed covariates and the target across environments when provided with environment-specific information. This serves as a motivational example that emphasizes the need for OOD generalization methods that remain effective even when confounding support is limited and the confounder is continuous-valued.

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Minimizing residual: The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ quantifies how much additional information $\phi(\mathbf{X})$ provides about the true label Y beyond what is already contained in the prediction $\hat{Y}$. The residual term can be expressed as: $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ is related to the expected cross-entropy loss as: $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29]. Here $H(Y | \hat{Y})$ is known as the *refinement error* that measures the model’s ability to distinguish between classes. Minimizing the residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ thus involves two complementary strategies. First, one can directly minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ by making $\phi(\mathbf{X})$ conditionally independent of the target Y given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. Alternatively, one can minimize the refinement error $H(Y | \hat{Y})$, for instance, through training techniques that minimize refinement error for accurate predictions before minimizing the calibration error [10].

4.3 Impact of additional informative variables

Section 4.2 identifies the information terms that matter when the confounder is unobserved. In practice, datasets may contain proxies for the hidden confounder [4, 85, 62]. We therefore quantify how adding non-causal but informative proxy covariates changes each term in the decomposition.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

Since $U \rightarrow Y$, any covariate that is informative to U , is also informative to Y and vice-versa. From the predictive information decomposition in (1), to maximize $I(Y; \hat{Y})$, it is required to minimize the concept shift $I(Y; E | \phi(\mathbf{X}))$. Liu et al. [50] suggest based on their empirical analysis that collecting additional covariates $\mathbf{X}_I$ such that $\mathbb{P}(Y|\mathbf{X}, \mathbf{X}_I)$ is more stable across environments i.e., reducing concept shift, improves the OOD test accuracy. We theoretically show that utilizing more informative variables helps maximizing predictive information by maximizing or minimizing certain terms in (1).

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \quad & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) \quad & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) \quad I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

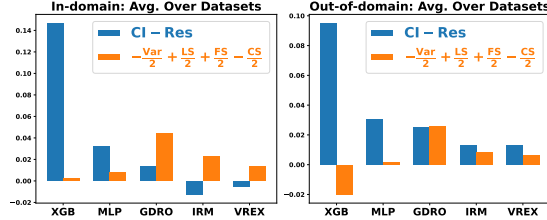
Proposition 4.3 shows that adding informative covariates increases conditional informativeness (5.i) and feature shift (5.ii), while reducing concept shift (5.iii). However, adding informative covariates also amplifies variation (5.iv), which may necessitate dedicated strategies to control the variation.

5 Experimental results

We conduct experiments on real-world and synthetic datasets to make three comparisons explicit: (i) whether real-world datasets exhibit nonzero shift metrics consistent with hidden confounding, (ii) whether the terms in (1) align with ID and OOD accuracy under hidden confounding, and

Table 2: Quantitative evidence of simultaneous shifts across eight real-world datasets. Mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples; one-sample t-tests against zero ($\alpha = 0.05$) confirm every reported shift metric is significantly nonzero (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: Method-level comparison of the predictive-information margin *conditional informativeness* – *residual* with average ID and OOD test accuracy. Larger margins align with higher accuracy; XGB has both the largest reported margin and the best average ID/OOD accuracy in the table.

(iii) whether adding informative covariates produces the predicted accuracy gains. The real-world evaluation uses eight tabular benchmark datasets from *TableShift* [24]: *Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*. We report ID-test and OOD-test accuracy for two ERM-based methods, XGBoost (XGB) [15] and multi-layer perceptron (MLP); two domain-generalization methods, IRM [6] and VREX [38]; and one robust-learning method, Group DRO (GDRO) [69]. Unless stated otherwise, each entry averages over 5 random hyperparameter choices as described in Appendix B; random-seed effects are statistically insignificant for these datasets [24]. Appendix § B gives the setup details, and Appendix C reports the corresponding hyperparameter-tuned results and dataset-specific decompositions.

Hidden confounding shift: Hidden confounding predicts simultaneous observable shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. The estimates below show this pattern across all eight real-world datasets: every reported label-shift, covariate-shift, conditional-covariate-shift, and concept-shift estimate is significantly different from zero. The nonzero values provide quantitative evidence that the benchmarks are not well described by a single isolated shift assumption. For qualitative corroboration, Appendix § E reports GPT-4o [1] queries listing plausible hidden confounders for benchmark datasets [24, 57]. In the *food stamps* dataset, for example, unmeasured state-specific *economic policies* may influence both *household income* and *food stamp reciprocity*.

Conditional informativeness vs. accuracy: Proposition 4.2 predicts that, under hidden confounding shift, higher predictive information is governed by the margin *conditional informativeness* – *residual*. Figure 4 reports this comparison for the first six datasets in Table 2, which have more than one training environment [57, 24]; dataset-wise plots are in Appendix C, and the formal sign-consistency comparison below also points to Appendix § C for dataset-level results. The visual method ranking is shown in Figure 1 and is consistent with the decomposition-based ranking. The empirical ordering follows Proposition 4.1: the margin is positively correlated with both ID and OOD accuracy, and XGB has the largest margin together with the highest average ID accuracy (85.22) and OOD accuracy (77.00). Dataset-wise results further show that the orange components are often close to zero because, under hidden confounding shift, $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ cancels as stated in Proposition 4.2. The relative ID/OOD ranking is also consistent with earlier benchmark findings [57, 24].

Informative covariates vs. accuracy: We next evaluate whether larger covariate sets move the terms in (1) in directions that coincide with accuracy gains. Following Nastl and Hardt [57], we compare

Table 3: Sign consistency links covariate-set expansions to accuracy improvements. Concept shift (**CS**) is consistently high across methods, while conditional informativeness (**CI**) is the favorable term emphasized by Proposition 4.2; XGB has the strongest CI value and the highest ID/OOD accuracies. **Res**: residual, **Var**: variation, **FS**: feature shift, **C**: causal, **AC**: arguably causal, **A**: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

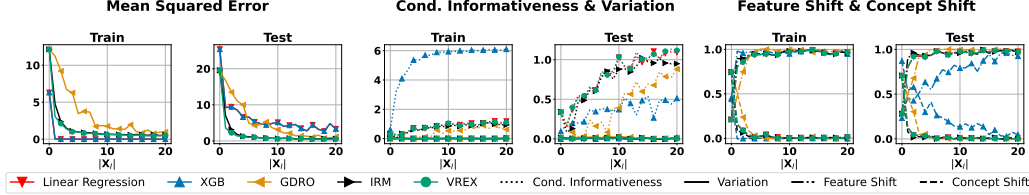


Figure 5: Synthetic comparison as the number of informative proxy variables $\mathbf{X}_I$ for U increases. More proxies reduce MSE, increase conditional informativeness and feature shift, and reduce concept shift, matching the predicted beneficial directions.

the nested subsets *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates* (Appendix § B). For each transition to a larger subset, the *sign consistency metric* records whether a term changes in its beneficial direction when accuracy improves: positively weighted terms such as conditional informativeness should increase, while negatively weighted terms such as concept shift should decrease. The metric is the fraction of such aligned changes for the terms in (1); Appendix § B gives the formal definition. Table 3 shows that conditional informativeness and concept shift have high sign consistency with accuracy improvements, with dataset-specific results in Appendix § C. The comparison is especially favorable for XGB: it achieves the highest conditional-informativeness sign consistency (0.89), ties for high concept-shift consistency (0.89), and has the best ID/OOD accuracies for all three covariate subsets.

Real-world datasets do not reveal whether *all available covariates* include *every relevant or sufficiently informative* variable. We therefore add a synthetic check with known causal structure $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ denotes informative covariates caused by the hidden confounder U . Figure 5 confirms the predicted pattern as more informative proxies are included: MSE decreases, conditional informativeness and feature shift increase, and concept shift decreases; the variation term remains zero because of the simplicity of the synthetic problem. Appendix § D gives the data-generation process and additional results.

6 Discussion, limitations, and future work

In summary, the predictive-information decomposition in Equation (1) and its hidden-confounding specialization in (1) provide a quantitative frame for explaining OOD performance under hidden confounding. The empirical evidence shows simultaneous nonzero covariate, label, conditional-covariate, and concept shifts across real-world benchmarks, while the method comparisons show that higher *conditional informativeness* – *residual* aligns with higher ID and OOD accuracy. Because hidden confounding entangles the shift terms (inequalities in (2), (3)), treating covariate, label, concept, or feature shift in isolation misses the observed pattern. Equation (4) instead favors learning environment-specific input–output relationships in addition to invariant structure, and the covariate-set and synthetic comparisons show why informative proxies can yield practical gains. Future work should quantify these gains more directly by (i) developing algorithms that remain effective with limited confounder overlap, (ii) measuring the cost–accuracy trade-off of acquiring informative covariates, (iii) handling entangled shifts without untestable proxy assumptions, and (iv) modeling ambiguity from unobserved confounders as a first-class source of OOD uncertainty.

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664 Appendix

665 A Proofs of propositions

666 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 667 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 668 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

669 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 670 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

671 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 672 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 673 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 674 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

675 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

676 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

677 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 678 expansion instead of $\phi(\mathbf{X})$. $\square$

679 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 680 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 681 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

682 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 683 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

684 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 685 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

686 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 687 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 688 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

689 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

690 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

691 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

692 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 693 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 694 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

695 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 696 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

697 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

698 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

699 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

700 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 701 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

702 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 703 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

704 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
705 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
706 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
707 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

708 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
709 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
710 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
711 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
712 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

713 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

714 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

715 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
716 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
717 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
718 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
719 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

720 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
721 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
722 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
723 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
724 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

725 B Experimental setup

726 In this section, we detail our experimental setup. The impact of random seeds on the results is
727 statistically insignificant in many settings [24]. For the results in the main paper, we report the mean
728 and standard deviation across five random hyperparameter values (sampled from their respective
729 domains in Table B1) for each dataset-method combination. Since we evaluate models on ID test and
730 OOD test data, to evaluate mutual information terms, we use the nonparametric entropy estimation
731 toolbox [37, 77, 78]. Following [24], we evaluate mutual information using both in-distribution (ID)
732 test data (from training domains) and out-of-distribution (OOD) test data (from test domains).

733 For consistency across datasets, we draw 20,000
734 random samples from each ID and OOD test
735 split when available; datasets with fewer than
736 20,000 samples use all available samples. The
737 evaluated mutual-information terms are not ma-
738 terially driven by this choice: Figure B1 shows
739 that the Equation (1) terms for XGB repre-
740 sentations on the *income* dataset remain relatively
741 stable as sample size varies. For neural models,
742 $\phi(\mathbf{X})$ is extracted from the layer immediately
743 before the classification head; for XGBoost,
744 the representation combines output margins and
745 SHAP values. All experiments were run on a single NVIDIA RTX A6000 GPU.

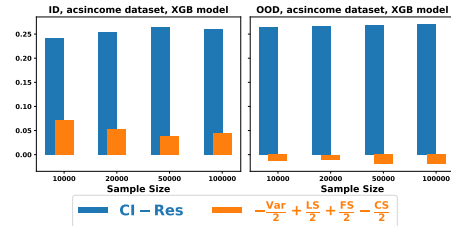


Figure B1: Sample-size sensitivity check for the predictive-information terms on the *income* dataset; the reported terms remain relatively stable across sample sizes.

746 To evaluate how covariate-set size changes the decomposition terms, we use the categories from Nasti
747 and Hardt [57]: *causal*, *arguably causal*, and *all covariates*. Causal covariates have a well-supported
748 direction of influence on the target and unlikely reverse causation, but restricting to them can omit
749 relevant parents when domain knowledge is incomplete. Arguably causal covariates include variables
750 with known causal status, plausible bidirectional influence, or likely but unconfirmed causal effects.
751 These groups approximate true causal parents using available knowledge, while allowing for possible
752 confounding. Some datasets also include *anti-causal* covariates, where the target likely affects the co-
753 variate but not conversely. The *all covariates* set includes every observed variable regardless of causal

Table B1: Hyperparameter grids used for the model comparisons reported in the paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

status. The sign consistency metric from the main paper § 5 records whether each term moves in its accuracy-improving direction when moving to larger covariate sets. For each $(m, \sigma_m) \in M$ where $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1), (\text{concept shift}, -1), \text{ and } (\text{residual}, -1)\}$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

Here P is the set of ordered covariate-set pairs (s, t) with $t \supset s$; for example, s may be causal covariates and t may be arguably causal covariates. The denominator includes the factor of 2 to count both train and test measurements. $\mu_s^{\text{tr/te}}(m)$ denotes the mean of measure m for covariate set s

on training/testing data, and $\mathbb{I}(\cdot)$ is the indicator. For the three covariate settings $\{causal, arguably causal, all\}$, $|P| = 3$. The final sign consistency value for term m is averaged across datasets as $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$, where $|D|$ is the number of datasets.

C Additional results on real-world datasets

This section expands the real-world comparisons behind the main paper. Figures C2 and C3 report, for each dataset, the two components aggregated in Figure 4: *conditional informativeness – residual* and *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2*. Proposition 4.1 predicts that their sum should align with model performance, and the plots show that the blue plus orange total is positively correlated with performance. Under the hidden-confounding patterns present in these datasets, the *conditional informativeness – residual* component contributes more to overall predictive information than the combined shift component, reinforcing the main-paper emphasis on conditional informativeness.

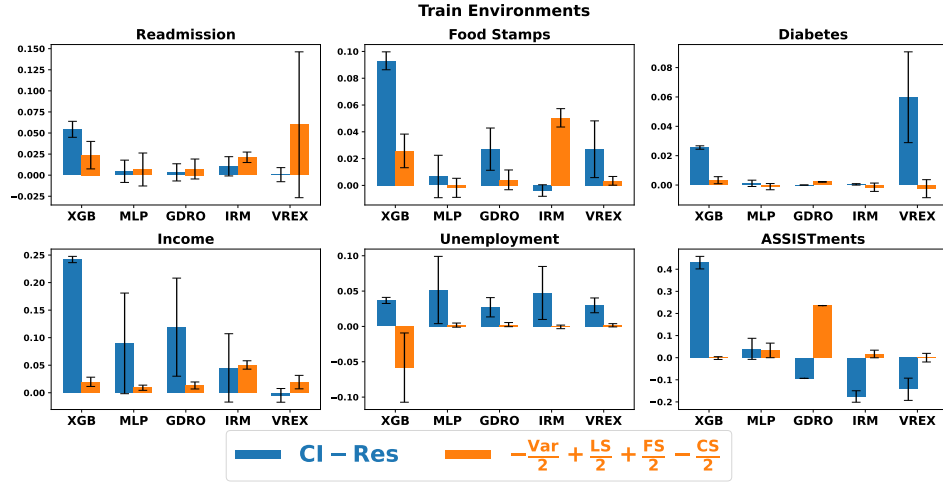


Figure C2: Dataset-specific decomposition results on ID data.

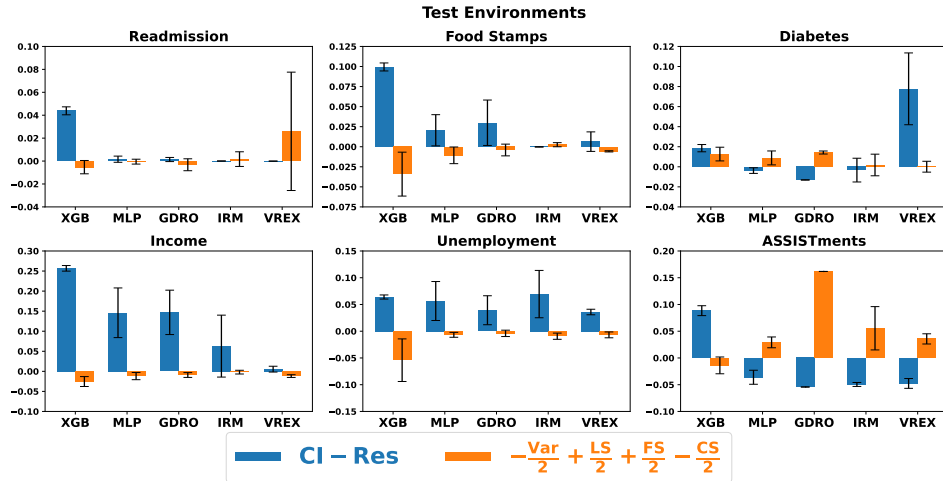


Figure C3: Dataset-specific decomposition results on OOD data.

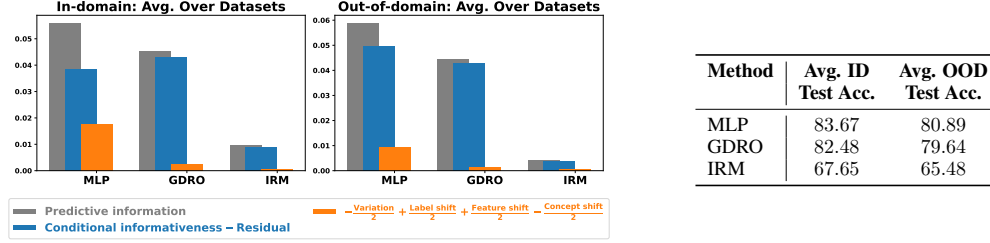


Figure C4: The difference *conditional informativeness* – *residual* in the plots is positively correlated with the average ID and OOD test accuracy over five datasets shown in the table on the right. The contribution of the difference *conditional informativeness* – *residual* is higher towards the overall predictive information when compared to the contribution of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

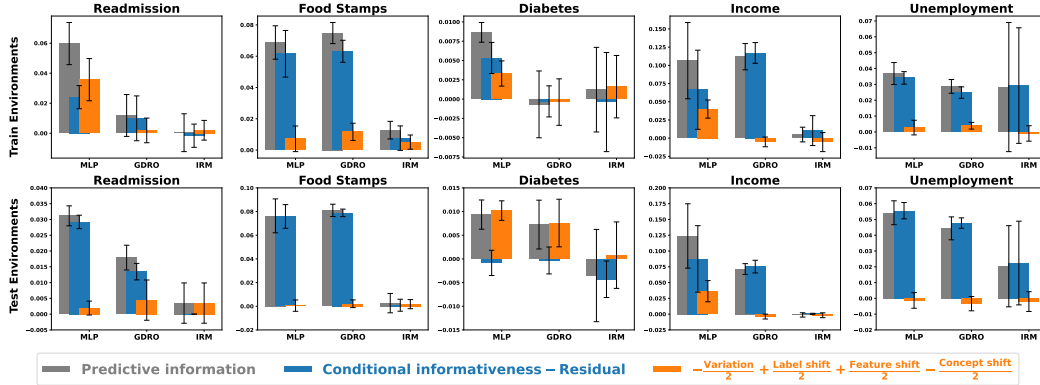


Figure C5: Dataset-specific comparison of predictive-information terms.

Table C2: ID and OOD test accuracy comparison across models.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9
	Income		Unemployment			
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.		
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05		
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00		
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00		

Figures C6- C15 further compare the decomposition terms in (1) for every method, dataset, and covariate subset, using Equation (1) as the shared accounting identity. The plots make the cancellation pattern visible: the *variation* and *feature shift* columns are similar in magnitude, so their signed difference contributes little to the total decomposition; likewise, *label shift* and *concept shift* are often similar in magnitude. Across nearly all settings, the margin *conditional informativeness* – *residual* is the largest contributor to predictive information, reinforcing the accuracy patterns reported in the main text.

Hyperparameter tuning: We also report a tuned comparison in which hyperparameters are selected by ID validation accuracy, not OOD validation accuracy. The grid search covers 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from the grid), and each selected experiment is rerun for 5 random seeds to estimate means and standard deviations for accuracy and information terms. Figure C4 preserves the main empirical pattern after tuning: the sum of *blue* and *orange* terms is positively correlated with model performance, and the

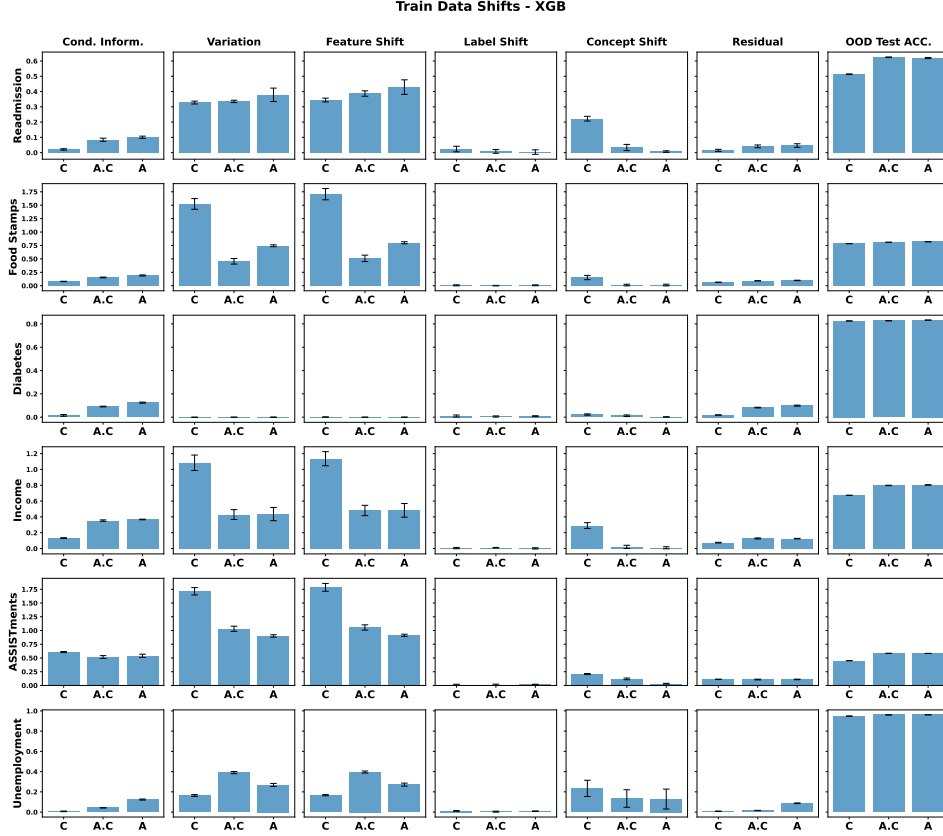


Figure C6: Train-set decomposition metrics for XGBoost across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

786 *conditional informativeness* – *residual* component contributes more than the combined shift component – *variation* / 2 + *label shift* / 2 + *feature shift* / 2 – *concept shift* / 2. Figure C5 and Table C2
787 show the same trend at the dataset level.
788

789 **Potential limitations with predictive information decomposition analysis:** Predictive information
790 $I(Y; \hat{Y})$ is useful for explaining the observed accuracy patterns, but it measures statistical dependence
791 rather than accuracy itself. For instance, a binary classifier that systematically predicts the *wrong*
792 label (e.g., $\hat{Y} = 1$ when $Y = 0$) achieves maximal $I(Y; \hat{Y}) = 1$ bit (for binary variables) due to
793 perfect anti-correlation, yet yields 0% accuracy. This occurs because mutual information quantifies
794 *reduction in uncertainty* rather than correctness. Other decomposition terms provide further insight:
795 conditional informativeness $I(\phi(X); Y|E)$ may mask harmful variation $I(\phi(X); E|Y)$, while feature
796 shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ through spurious correlations. Nevertheless, this decomposition
797 remains valuable for analyzing reasonably performant models, as our experiments demonstrate - it
798 reveals whether predictive information stems from generalizable patterns or environment-specific
799 artifacts, enabling targeted improvements while maintaining interpretability.

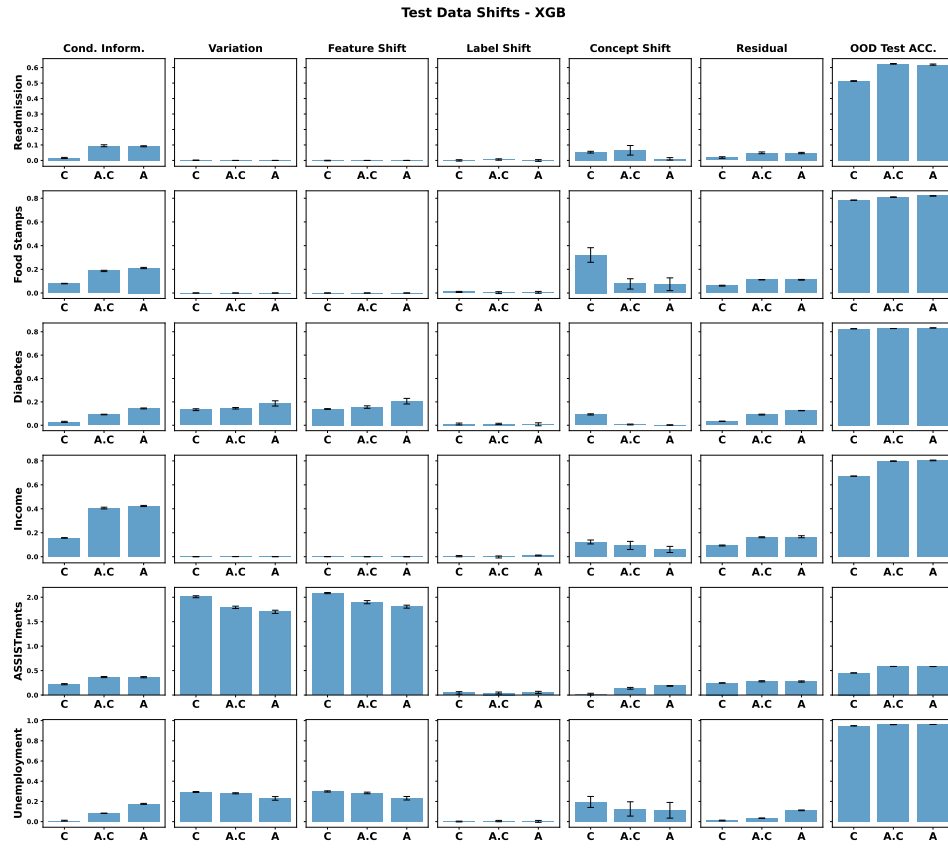


Figure C7: Test-set decomposition metrics for XGBoost across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

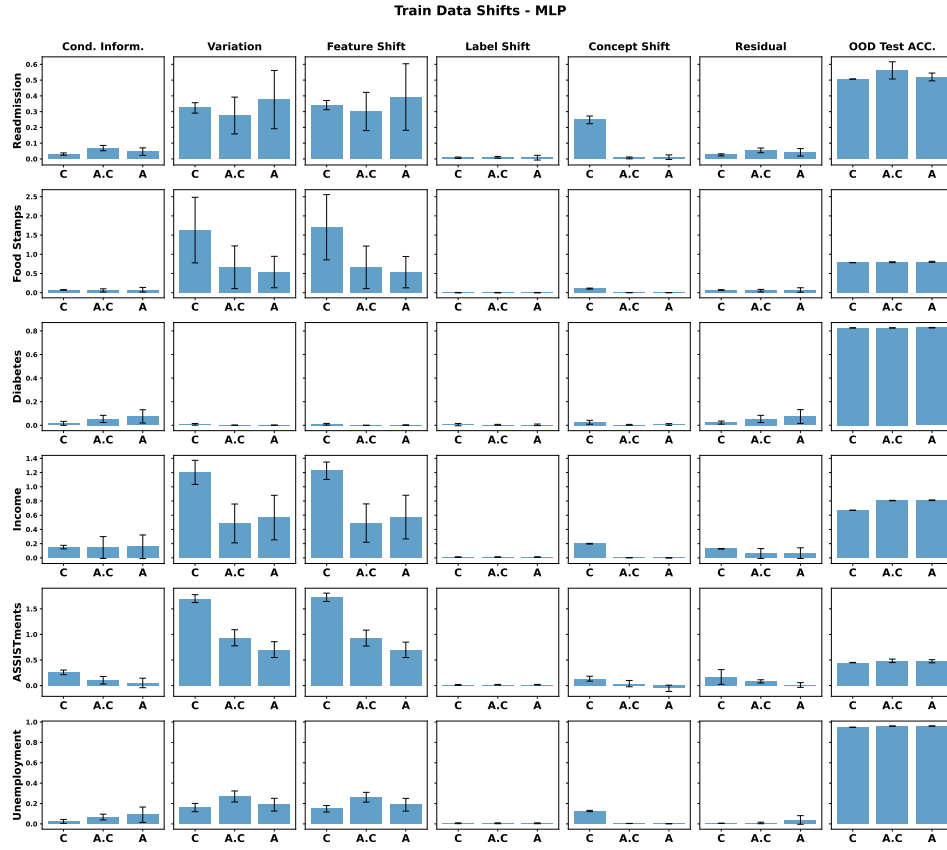


Figure C8: Train-set decomposition metrics for MLP across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

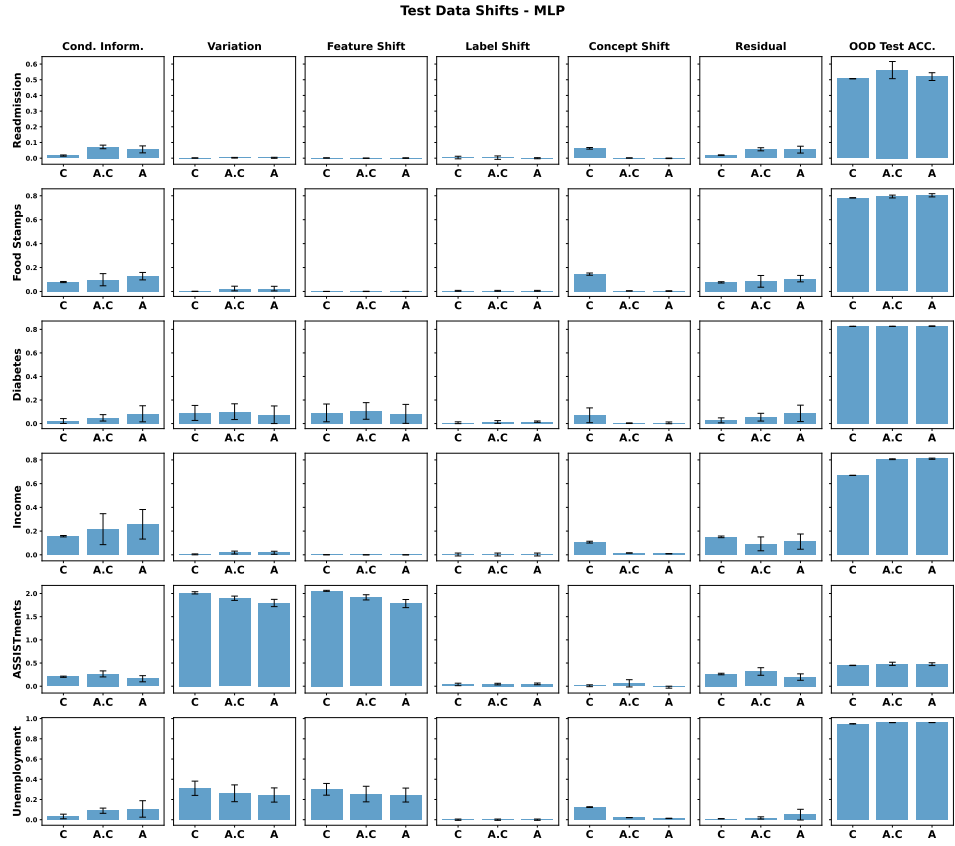


Figure C9: Test-set decomposition metrics for MLP across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

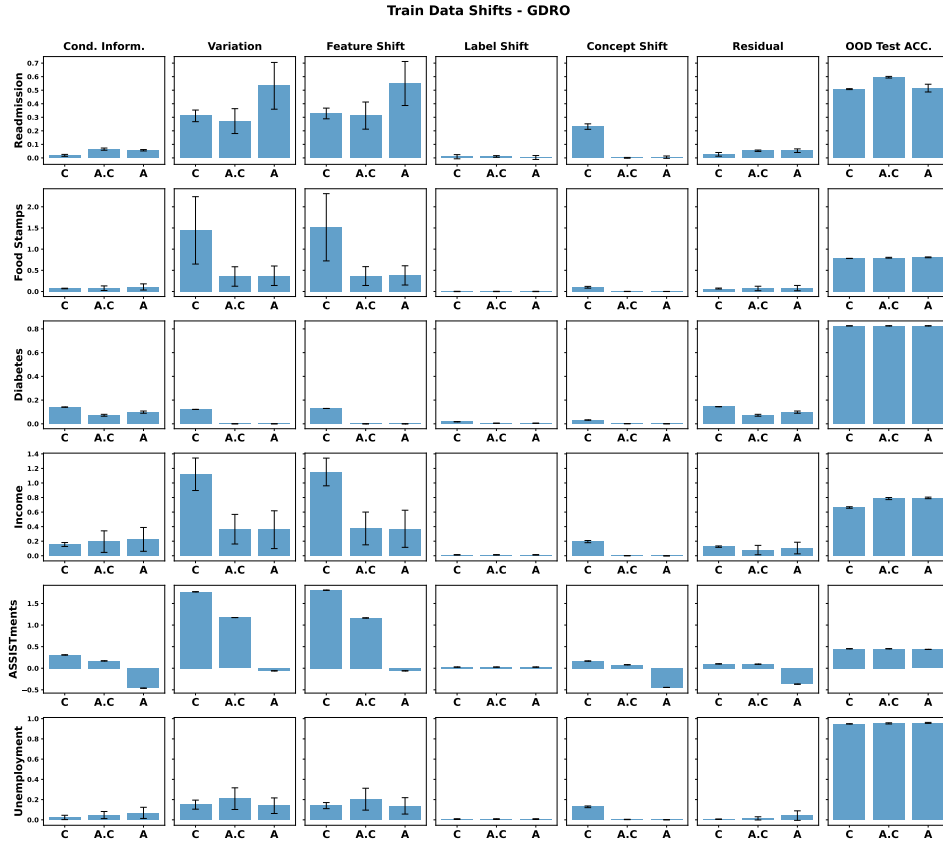


Figure C10: Train-set decomposition metrics for GDRO across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

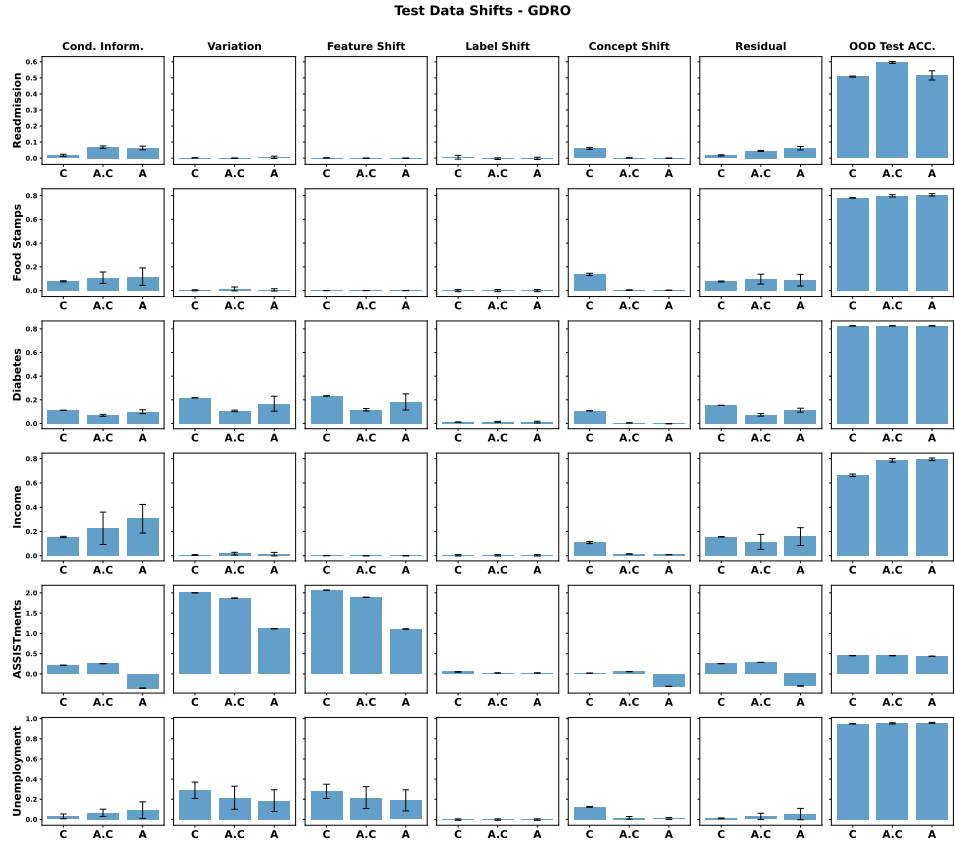


Figure C11: Test-set decomposition metrics for GDRO across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

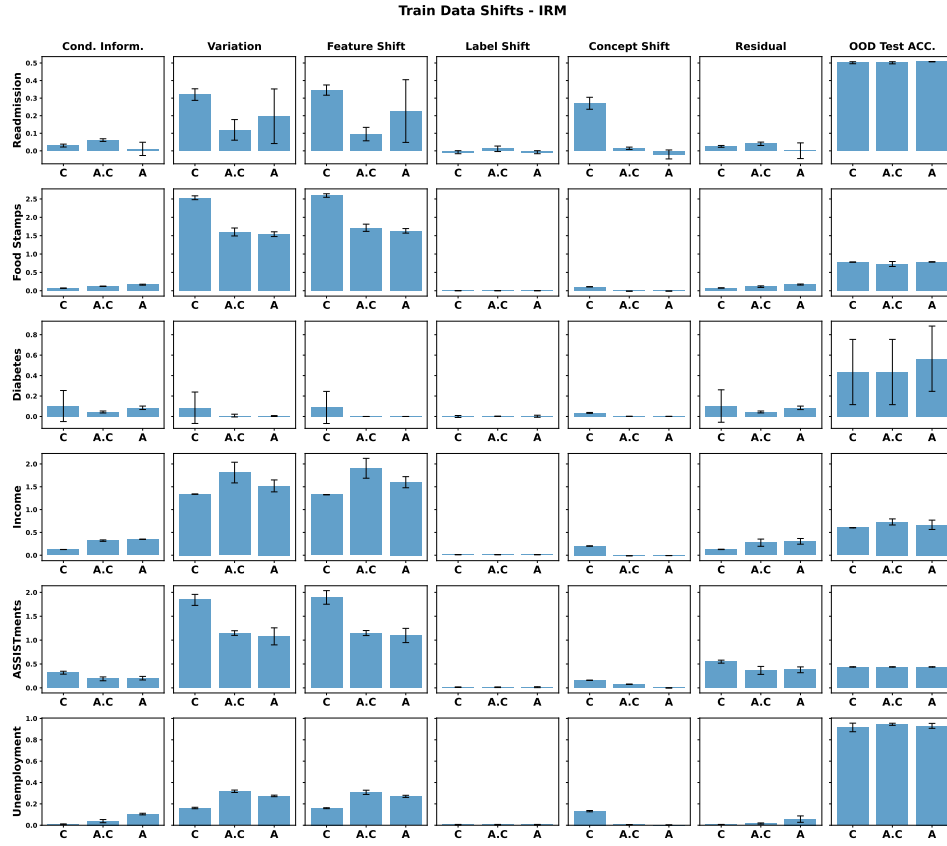


Figure C12: Train-set decomposition metrics for IRM across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

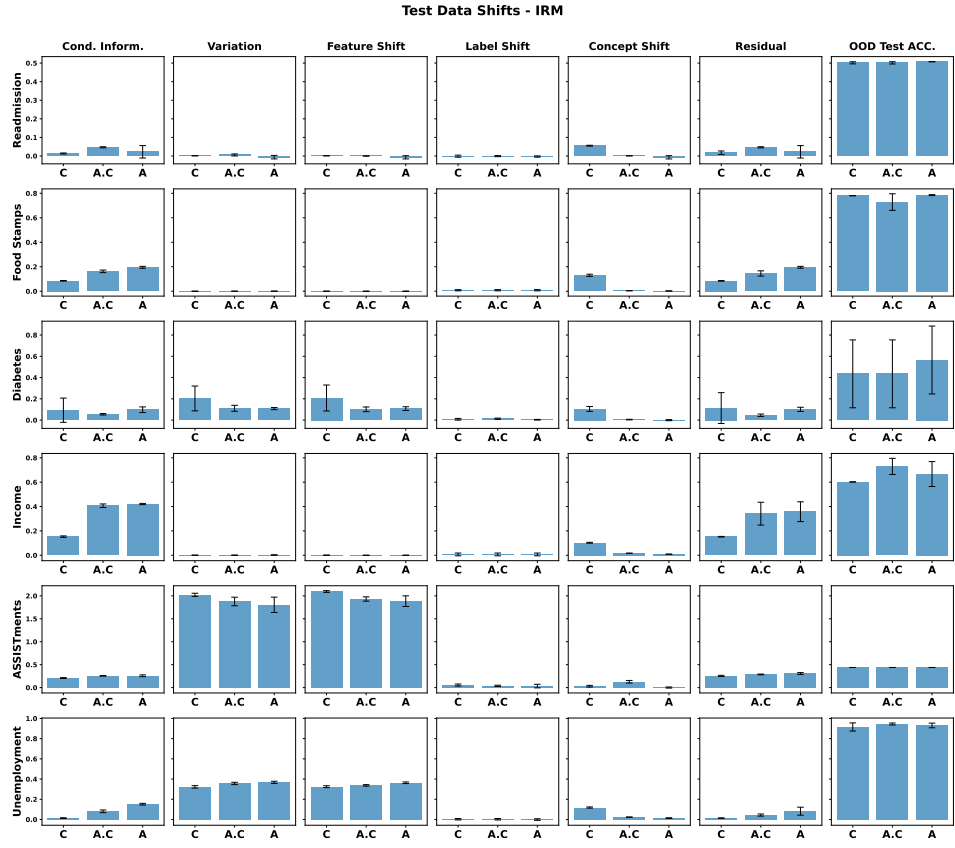


Figure C13: Test-set decomposition metrics for IRM across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

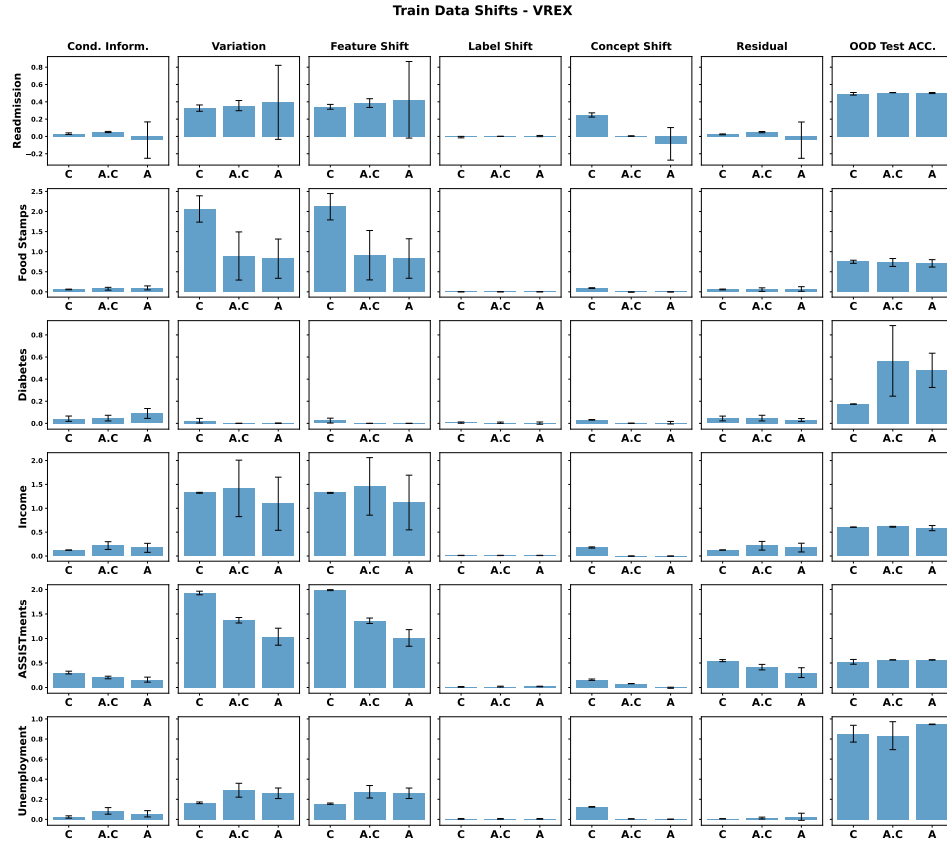


Figure C14: Train-set decomposition metrics for VREX across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

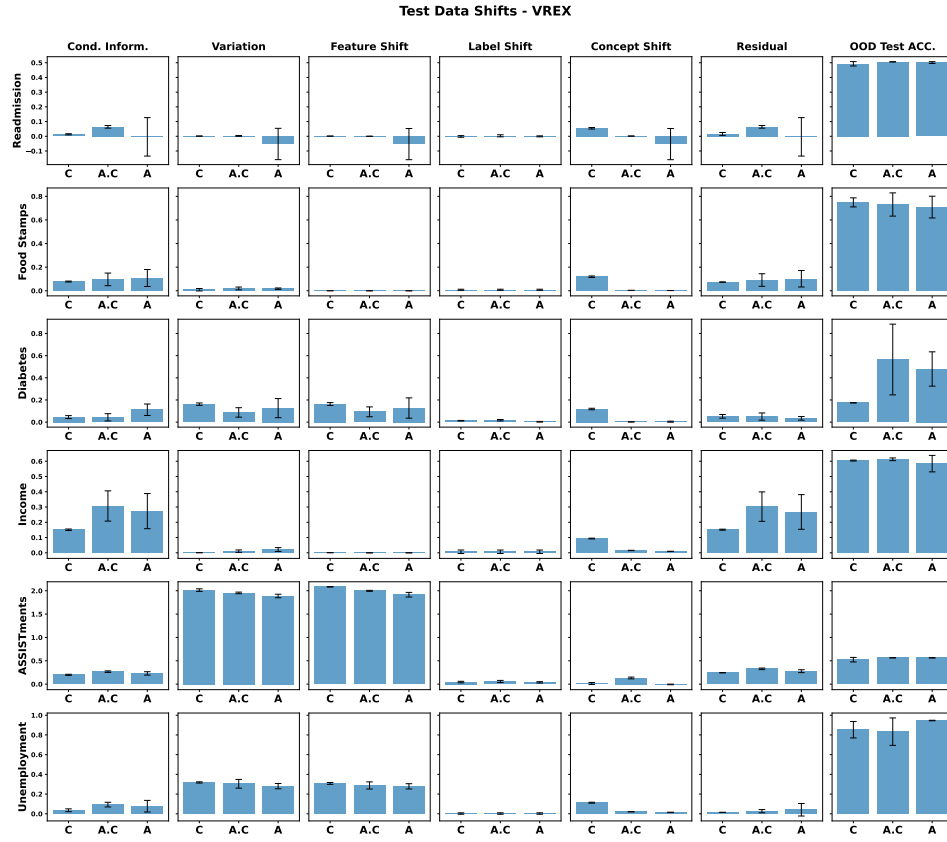


Figure C15: Test-set decomposition metrics for VREX across feature groups: (C) Causal, (A.C) Arguably causal, (A) All.

800 D Results on synthetic datasets

801 The synthetic experiments isolate the quantitative mechanisms behind the real-world patterns by
 802 varying (i) informativeness criteria for proxy variables, (ii) confounder overlap across environments,
 803 and (iii) the number of causal versus anti-causal variables. The data-generating process follows (12)
 804 with causal structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, where U is unobserved, X is
 805 observed, Y is the target, and $\mathbf{X}_I$ contains additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

806 We evaluate multiple functional forms for f_X, f_i , and f_Y . Domain shifts are induced by changing
 807 the environment-specific confounder mean μ_u^e , allowing us to compare model performance across
 808 different degrees of distributional shift.

809 **Informativeness vs. accuracy:** In this set of experiments, we begin with linear structural equations:
 810 $f_X = 0.3U, f_i = 0.1U$, and $f_Y = X - 2U$. For these experiments $\mu_u^e \in \{-2, 2\}$ for training
 811 environments and $\mu_u^e \in \{0, 4\}$ for test environments. For the results presented in Figure 5 of the
 812 main paper, we consider one hidden confounding variable with $|\mathbf{X}_I| = 20$ informative covariates.
 813 As we increase the number of informative covariates from 0 to 20, we observe: (i) a reduction in
 814 mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while
 815 (iv) decreased concept shift. We extend these findings by examining two additional informativeness
 816 criteria. First, we analyze the case with one hidden confounder and a single informative covariate
 817 ($|\mathbf{X}_I| = 1$) across varying noise levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$. Figure D16 demonstrates that as noise
 818 decreases from 2.0 to 0.0, we observe: (i) reduced mean squared error, (ii) improved conditional
 819 informativeness, (iii) enhanced feature shift, alongside (iv) decreased concept shift. Consistent with
 820 other synthetic experiments, the variation term remains zero throughout. Finally, we investigate a
 821 setting with 20 hidden confounders and corresponding 20 informative covariates, using modified
 822 structural equations $f_X = 0.3U, f_i = 0.2U$, and $f_Y = X - 2U$. Figure D17 shows that as we
 823 introduce informative covariates corresponding to hidden confounders, we again observe: (i) reduced
 824 mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, while
 825 (iv) decreased concept shift. These experiments demonstrate how different informativeness criteria
 826 help in achieving OOD generalization.

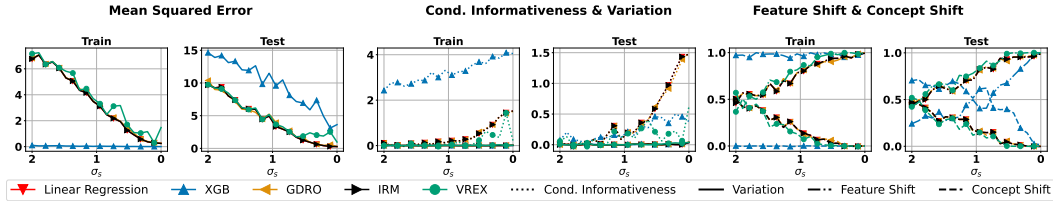


Figure D16: Lower-noise proxy variables provide stronger quantitative gains: MSE decreases, conditional informativeness and feature shift increase, and concept shift decreases.

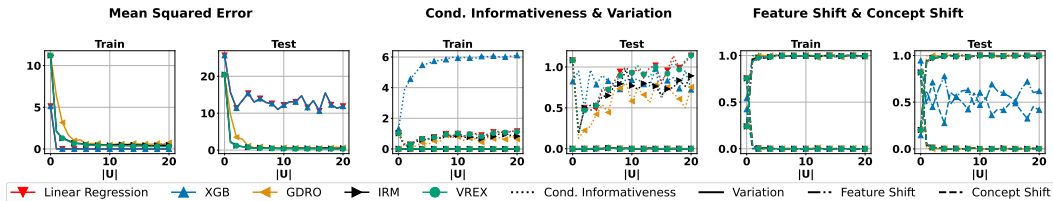


Figure D17: Adding more informative proxy variables $\mathbf{X}_I$ for hidden confounders U reduces MSE, increases conditional informativeness and feature shift, and reduces concept shift.

827 **Overlapping confounder support vs. accuracy:** As established in the main paper, recent work
 828 by Prashant et al. [62] proposed an OOD generalization method for hidden confounding shift that
 829 assumes test confounder support remains within training support. While our results demonstrated

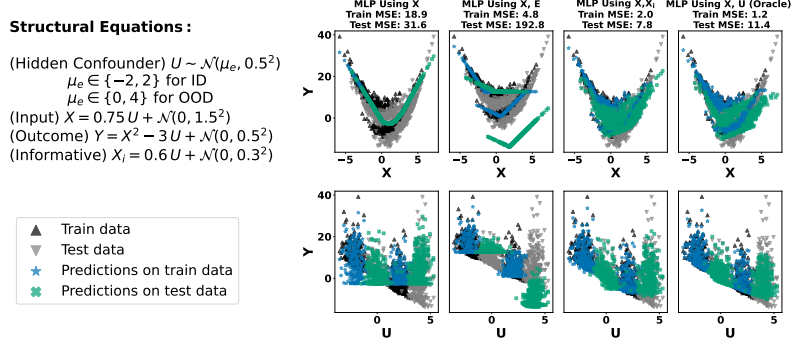


Figure D18: Performance of MLP in the low-overlap non-linear setting, comparing feature sets: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

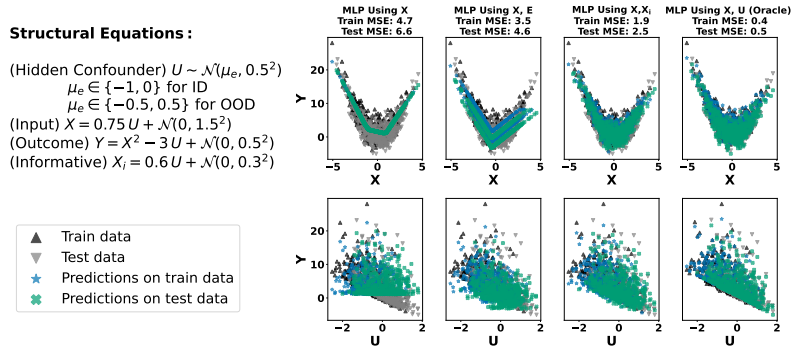


Figure D19: Performance of MLP in the high-overlap non-linear setting, comparing feature sets: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

successful learning of correct relationships in synthetic linear data when sufficient information about hidden confounders exists in observed covariates, we now extend this analysis to nonlinear data using MLP and XGB models under both low and high confounding support conditions (Figures D18–D21). In low confounding overlap settings (characterized by distant μ_e between ID and OOD data), models relying solely on X learn an inadequate global function that fails to distinguish environments. Performance improves when incorporating environment-specific summary statistics of observed covariates, which helps capture environment-specific relationships, but the most significant gains occur only when leveraging additional informative covariates, underscoring their importance. For completeness, we include both oracle model results and U - Y relationship scatter plots to elucidate the learned input-output mappings. These experiments confirm better performance under high confounding overlap as expected, but more importantly, reveal promising results in the low-overlap regime - a previously unstudied scenario that challenges the common support assumption in the literature.

Number of causal vs anti-causal variables: Recent benchmarks [24] show that baseline methods (e.g., XGB, MLP) consistently match or outperform OOD generalization techniques (e.g., GDRO, IRM, VREX) in real-world datasets, coinciding with the prevalent causal structure $\mathbf{X} \rightarrow Y$ in tabular data. To test if baseline superiority stems from the causal structure: $\mathbf{X} \rightarrow Y$, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (e.g., $\rho = 0.5$ gives $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). Results (Figure D22) show baseline resilience even under the causal structures $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, though no method dominates universally (Figures D23, D24). Baselines achieve superior PMA-OOD scores [24] (fraction of maximum OOD accuracy). We evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76], TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-REx [38], Domain Mixup [90, 91]); and (3) Robust optimization methods (DRO [42], GroupDRO [69], Label DRO, Adversarial Label DRO [94]).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

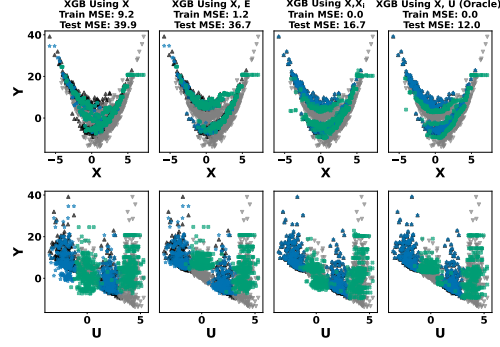


Figure D20: Performance of XGB in the low-overlap non-linear setting, comparing feature sets: (i) X (ii) X and E (environment statistics) (iii) X and X_i (iv) X and U (Oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

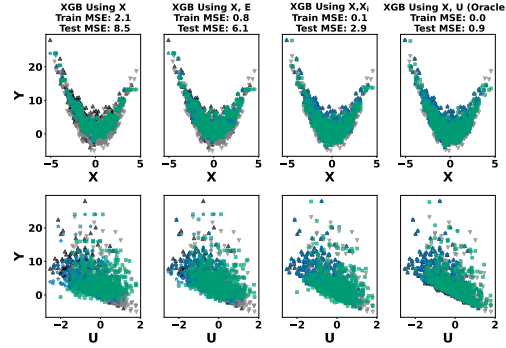


Figure D21: Performance of XGB on the high overlap non-linear setting, for feature sets: (i) X , (ii) X and E (environment statistics), (iii) X and X_i , and (iv) X and U (Oracle).

Percentage of Max Accuracy (PMA-OOD) on Synthetic Datasets Across All ratios ($\rho = \frac{|X_c|}{|X_t|}$)
Left: Data With Linear Equations, Right: Data With Nonlinear Equations

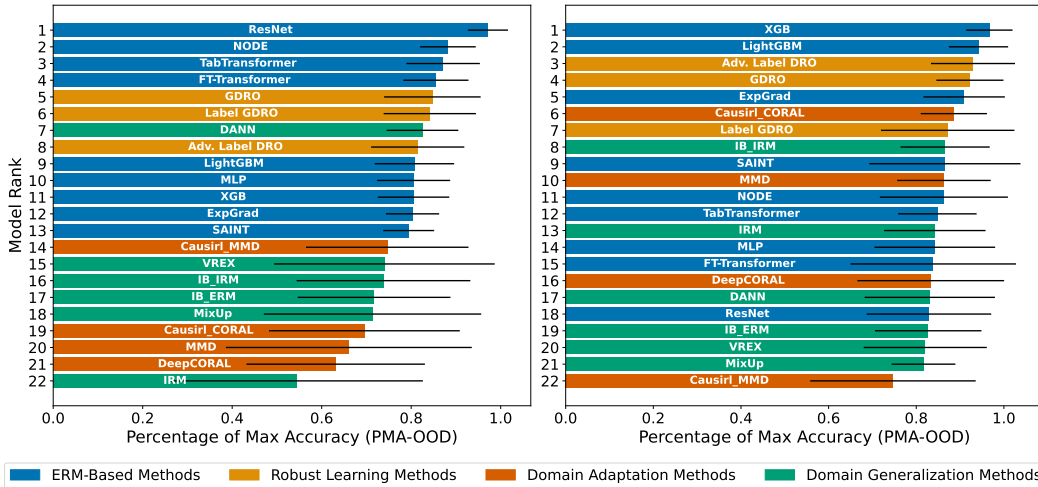
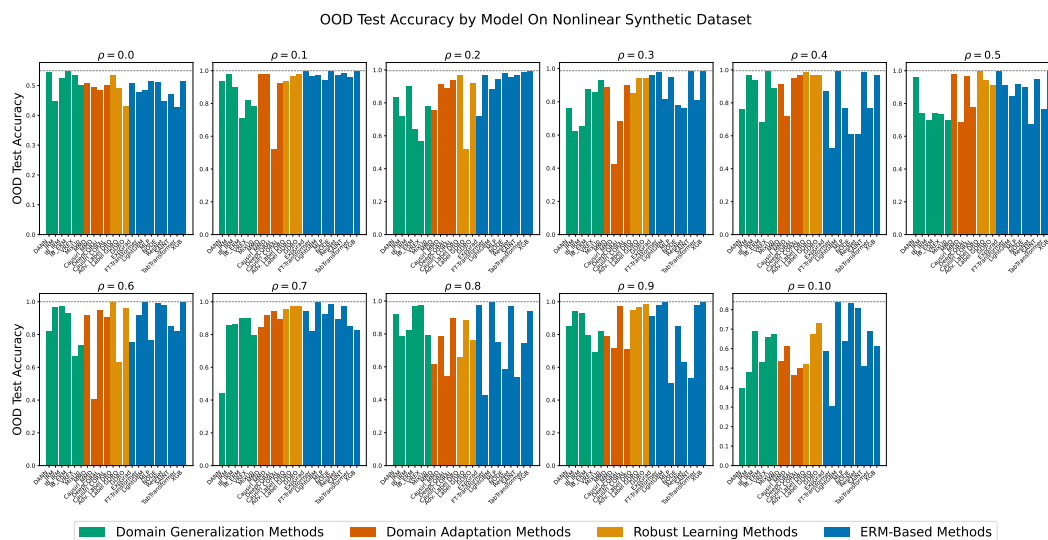
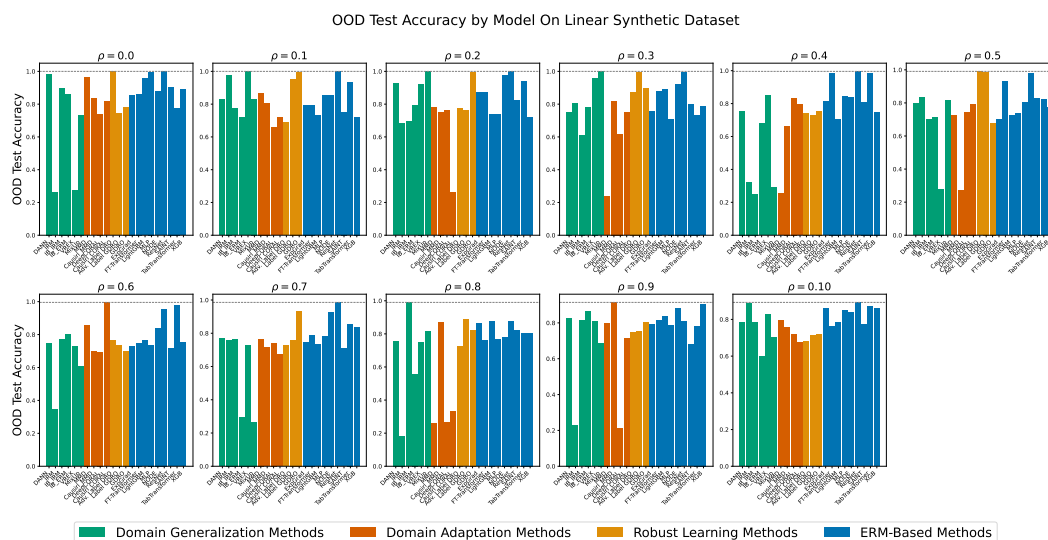


Figure D22: Synthetic PMA-OOD comparison: ERM-based methods achieve a higher percentage of maximum OOD test accuracy, consistent with the real-world pattern in [24].



E Qualitative analysis of hidden confounding shift in real-world datasets

This section provides qualitative support for the quantitative shift evidence in the main paper. Tables E3-E18 list plausible hidden confounders that could influence observed covariates in real-world datasets [57]. For dataset details, we refer to [24, 57]. We use GPT4o [1] for this task and do not treat the generated lists as exhaustive. Instead, the tables illustrate why the nonzero shift metrics in Table 2 are practically plausible across domains.

Prompt: We use the prompt below to query GPT4o for plausible hidden confounders.

For a target variable, you will be given lists of causal, arguably causal, anti-causal, and spurious covariates. Also, you will be provided with environment variables such that different datasets are collected in environments induced by these environment variables. Provide possible hidden confounders that cause any of the target, causal, arguably causal, spurious, and anti-causal covariates. That is, the hidden confounders should influence the distribution of their children in different domains. Ensure hidden confounders are not a part of the given variables, and your explanation should include how hidden confounders influence their children in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.